



Does medieval trade still matter? Historical trade centers, agglomeration and contemporary economic development[☆]

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ABSTRACT

This study establishes a link between medieval trade, agglomeration and contemporary regional development in ten European countries. It documents a statistically and economically significant positive relationship between prominent involvement in medieval trade and regional economic development today. The analysis indicates that a long-lasting effect of medieval trade on contemporary regional development is transmitted via its effect on agglomeration and industry concentration. Further empirical analyses show that medieval trade positively influenced city development both during the medieval period and in the long run; they also reveal a robust connection between medieval city growth and contemporary regional agglomeration and industry concentration. This research thus highlights the long-run importance of medieval trade in shaping the development of cities as well as the contemporary spatial distribution of economic activity throughout Europe.

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1. Introduction

There is ample evidence that trade is an important determinant of both long- and short-run economic development. However, most of the existing literature focuses on the impact of 19th-century trade on market integration or the “Great Divergence” (e.g., [Galor and Mountford, 2008](#); [O'Rourke and Williamson, 2002](#); [Pascali, 2013](#)), or on the impact of contemporary, Post-World War II trade activities on recent economic growth and development performance across countries ([Dollar and Kraay, 2003](#); [Frankel and Romer, 1999](#)). There are few studies (e.g., [Acemoglu et al., 2005](#)) considering the effect of cross country trade in periods earlier than the 19th century. In this study, the authors investigate the impact of early modern long-distance overseas trade for institutional developments and the pre-industrial development process across European countries.

Hence, until now there has been no study exploring the long-lasting effects of trade and commerce in European cities during the

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High and Late Middle Ages. The importance of medieval trade for the development of cities and regions in the Middle Ages and the following centuries is widely accepted. Apart from this, no research has acknowledged the fact that medieval trade might also have long-term influences on regional development persisting until today; this despite the fact that medieval trade, through its potential impact on agglomeration and spatial concentration of industry, could have led to path-dependent regional development processes resulting in developmental differences surviving over the centuries.

The aim of this study is to provide evidence that medieval trade, as a result of its impact on agglomeration, has caused differences in regional development that remain visible today. It therefore provides a new explanation for the uneven distribution of economic activity and significant spatial concentration of industries throughout Europe (e.g., [Chasco et al., 2012](#); [Koh and Riedel, 2012](#); [Roos, 2005](#)). It also contributes to the understanding of the still puzzling persistent differences in regional economic development ([Becker et al., 2010](#); [Maseland, 2012](#) or [Tabellini, 2010](#)). Furthermore, the study offers an additional explanation for the rapid central European urbanization process that began during the Middle Ages, has continued until today and led to the so called “Rise of the West” ([Bosker et al., 2013](#)). Why was the urban development in central Europe since the Middle Ages not characterized by a reversal of fortunes and periods of stagnation like that in the Near and Middle East? [Bosker et al. \(2013\)](#) highlight the role of local participative political institutions. I aim to show that medieval commercial activities also constitute an important cornerstone of this development. Finally, this study contributes to a

growing literature reporting on the persistence and path-dependent nature of spatial equilibria and city growth processes (Bosker et al., 2007; Bleakley and Lin, 2012; Davis and Weinstein, 2002; Davis and Weinstein, 2008; Michaels and Rauch, 2014; Miguel and Roland, 2011 and Redding et al., 2011).

To establish a link between medieval trade, agglomeration and contemporary performance I link the typical characteristics of medieval trade and cities to the determinants of agglomeration suggested by New Economic Geography (NEG) and agglomeration economics (e.g., Glaeser et al., 1992; Krugman, 1991). In a second step, based on studies combining NEG, endogenous growth models, and the theory of path-dependence, I propose a positive connection between agglomeration, industrial concentration and contemporary development. The underlying idea is the following: While medieval trade activities clearly reflect the effect of the determinants of medieval trade like e.g. location fundamentals and second nature geography on development, I argue that trade activities themselves had a significant additional effect through their direct influence on agglomeration patterns.

Afterwards, I test the causal chain from medieval trade through agglomeration and on to contemporary regional economic development by using rich regional and city level data sets and a wide range of empirical methods. In this empirical investigation I use three variables to capture medieval trade activities. First, based on several historical sources and trade route maps, I construct a dummy variable identifying cities that were important centers of trade during the medieval period. Second, I calculate a variable that shows the distance between each region or city and the closest of these trade cities. This variable allows me to test whether trade activities lead to the emergence of spatial core-periphery patterns as implied by theoretical considerations. Third, I compute a variable reporting the number of centuries a city can be considered to be an important trade city.

The results of the empirical estimations provide strong evidence for a significant relationship between medieval trade and contemporary regional economic performance. Furthermore, a detailed empirical investigation on city level shows that medieval trade activities are robustly positively associated with city development both during the medieval period and in the long run. Therefore, the observed path-dependent development process of European cities is partly rooted in the persistent effect of medieval trade activities. Moreover, I also find that the effect of medieval trade on contemporary regional development can be explained by its influence on agglomeration patterns. This is shown by the fact that medieval trade activities are strong direct predictors of today's spatial distribution of economic activity.

Importantly, I show that the results are robust to the inclusion of many geographic, political, economic and historical covariates of development and medieval trade, as well as different samples, data sets and medieval trade measures. I also show that a bias arising from unobserved heterogeneity is unlikely as the results hold also when considering only regions that were historical urban centers. Even among this homogeneous group, historical urban centers that were also hubs of medieval trade show a higher contemporary GDP per capita than urban centers that were not important in medieval trade.

The remainder of the article proceeds as follows. First, I theoretically establish the link between medieval trade, agglomeration and present-day economic development. Afterwards, I introduce and discuss the most important variables and data and explain the empirical setting. Next, I conduct the empirical analysis and interpret and discuss the results in detail. Finally, I conclude and summarize the main findings.

2. Theoretical considerations and related literature

It is a well-established idea that trade was a decisive factor for the development of medieval cities and the revival of city

growth during the period of the so-called “Commercial Revolution” (e.g., Lopez, 1976; Cantoni and Yuchtman, 2014). History provides many examples of cities owing their importance primarily to their function as centers of trade, such as the German cities of Nuremberg (Nicholas, 1997), Frankfurt (Holtfrerich, 1999), Cologne (King, 1985) or the Polish city of Gdansk.¹

Using concepts developed by NEG (Krugman, 1991) and agglomeration economics, one can explain why medieval trade was important for the rise of cities in medieval Europe. This is achieved by linking the characteristics of medieval trade and trade cities to second nature (man-made) causes of agglomeration like knowledge spillovers (e.g., Glaeser et al., 1992; Henderson et al., 2001). In medieval times, the economy, especially the urban economy, was characterized by a high degree of regional specialization (e.g., Postan et al., 1952 or Pounds, 2005). For instance, the Southern German cities that became important trade centers in the later medieval era specialized in textiles (Barchent etc.) and paper production. The different regions exported what they specialized in—or had a comparative advantage in, e.g., due to natural resources—and imported what they did not have themselves. This specialization of trade cities in a particular industry or sector gave rise to the existence of technological (non-pecuniary) externalities like Marshall–Arrow–Romer (MAR) externalities (Marshall, 1890; Romer, 1986) or Porter externalities.²

A second important characteristic of medieval trade cities was the comparatively high variety of goods that were available. Those assortments of goods were available first at the local markets, then at the large trade fairs in the Champagne region and other important trade cities (such as Frankfurt, Cologne, Ulm, etc.), and then, in the late medieval age, in the branches and kontors of the Hanseatic League and trading companies (“super-companies”) like the Fugger in Augsburg.

The latter two in particular also supplied luxury goods and exotic commodities from the Far East, as long-distance trade was reestablished at the beginning of the Late Middle Ages. I can consider this high variety of goods as an important demand-side driven agglomeration force, as it makes a city more attractive to settle in.

Additionally, the large variety of goods and prospering industry gave rise to the self-reinforcing circular causation caused by backward and forward linkages and lead to agglomeration and core-periphery patterns in NEG models (Krugman, 1991; Ottaviano and Thisse, 2004). These forward and backward linkages constitute the virtuous circle that generates agglomeration and uneven spatial distribution of population and economic activity.³

However, the main argument of this paper is that medieval trade had significant consequences for economic development today. Reassuringly, the self-reinforcing nature of the described agglomeration and concentration processes implies a path-dependent process of city development. This path-dependent development process resulted in differences in concentration of economic activity and population that remain evident today. Cities that were involved in medieval trade activities over a sufficient period of time became locked onto a superior development path by comparison to other cities without that history. This is a typical characteristic of processes caused by increasing returns or positive feedback (David, 2007). Several studies (e.g., Bosker et al., 2007 and Davis and Weinstein, 2002, 2008)

¹ Obviously, there are exceptions to this story, i.e. cities and regions becoming large and important agglomerations without having been important centers of medieval trade. Some of these exceptions are discussed in detail in Appendix C.1.

² Nicholas (1997) additionally points to the fact that over the course of the Middle Ages the industry dominating in a city, e.g. the textile industry, increasingly diversified. This intra-industry diversification could be an additional channel through which technological externalities could have arisen.

³ For these arguments to hold, it is crucial that the medieval markets showed a certain degree of integration and that I actually observed a urban-rural wage differential. Evidence on these aspects is provided in Appendix C.2.1.

show that city growth is characterized by a long-run persistence that is immune even to such shocks as the Second World War. Thus, there is a fair amount of empirical evidence pointing towards the path-dependent character of agglomeration processes and city development. In addition, there are numerous examples of historical events and phenomena with long-run impacts on economic development (e.g., Acemoglu et al., 2001; Alesina et al., 2013; Comin et al., 2010; Nunn and Wantchekon, 2011). I argue that medieval trade can be added to this list of events.

If this reasoning is correct it would mean that medieval trade—through its effect on agglomeration patterns—affected development over and above the effect of location fundamentals and second nature geography that is also reflected in medieval trade activities. Furthermore, it would be in line with evidence found by Roos (2005) and the predictions from NEG. However, this contrasts with the papers of Ellison and Glaeser (1999) and Wolf (2009), which emphasize the significance of first nature causes of agglomeration. Furthermore, I trace back significant parts of today's differences in regional economic development to the differences in involvement in trade activities during the medieval period. Hence, if I find empirical support for my conjectures, this shows that historical events (like, e.g., the division of Germany or the Second World War) or political measures in more recent times are not the main determinants of regional economic prosperity—despite the fact that these events have influenced, for example, the location of airports in Germany (Redding et al. 2011). Instead, the theoretical reasoning and its empirical confirmation would support the findings of Maseland (2012) who demonstrates that regional development disparities in Germany are persistent and are largely explained by increasing differences between core areas and the periphery and not by an “east-west divide” originating from the division of Germany or other relatively recent population shocks (e.g. Schumann, 2014).

3. Data and setting

3.1. Setting and level of analysis

Because medieval trade took place in cities the empirical analysis should be based on city-level data. However, until now, communal and local level data is scarce and important measures (e.g., GDP per capita) are still not available on these level of aggregation. Hence, I base the empirical analysis on regional level data. I adhere to the NUTS (“Nomenclature of Units for Territorial Statistic”) regional classification, the official regional reference unit system used in the European Union (EU). I choose to conduct the analysis on the most disaggregated level for which the essential data (e.g., GDP per capita) is available. Therefore, I conduct the analysis with a NUTS-3 region as observational unit.⁴

3.2. Dependent variables and agglomeration measures

The dependent variable is the natural logarithm (ln) of GDP per capita in a NUTS-3 region, originating from the Eurostat regional statistics database. I take the latest available values from the year 2009. All other time-variant variables also come from the year 2009 to enable comparability.

As measure of spatial industry agglomeration I follow Roos (2005), Chasco et al. (2012) and others in using the ln of the relative

GDP density as measure for the spatial distribution of economic activity. The measure is calculated by dividing a region's share of a country's total GDP through its share of the country's total area. This means it shows whether the concentration of economic activity in a region is below or above the country's average. Additionally, I will also present results using population density as agglomeration measure. Table A.1 in the Data Appendix gives a descriptive overview of all variables used in the following empirical analysis.

3.3. Independent variables

This study aims to investigate the impact of trade between cities during the medieval age. To be able to identify the effect of medieval trade on agglomeration I collect data on important trade cities in Europe until 1500 CE. In the cross-sectional analysis I only consider the situation in 1500 CE, at the end of the medieval period while I take into account the date at which a city became a trade center in the subsequent panel data analysis.

The main sources of information on important medieval trade activities are maps printed in historical atlases (primarily Davies and Moorhouse, 2002; King, 1985; Magocsi, 2002 and Stier et al., 1956) and monographs (e.g., Spufford, 2002).⁵ I collect information about cities prominently involved in trade from four historical maps providing information about cities located on “major” or “important” trade routes in around 1500 CE (i.e., the late medieval period).

Because there is no consensus about the exact importance of trade cities and trade routes during the medieval period I consult several different sources to gather sufficiently reliable data. Furthermore, I use other qualitative information when judging the importance of the trade cities included. For example, I look at whether a city was an important member of the Hanseatic League or the capital of a quarter or a third (like, e.g. Dortmund or Cologne).⁶ Information on this is provided by Dollinger, (1966). Additionally, especially for less prominent trade cities (Paderborn, Soest, Harfleur, Tarent etc.), I also look at whether they were situated along well-known trade routes like the “Hellweg” in Germany (as is the case, e.g., for Soest). Moreover, I consult several standard historical sources on medieval trade activities in different Central European regions (e.g., Dietze, 1923, Hunt and Murray, 1999; Schulte, 1923; Spufford, 2002) and look at whether they mention a city as being prominently involved in trade or as having over-regional importance as a market, fair, or trading city. Finally, I also draw on other historical atlases—such as that of Kinder and Hilgemann (1970)—and other regional trade route maps (e.g., Schulte, 1923) as sources for validating the information in the primary maps.⁷ In the Data Appendix I report and discuss all these sources and provide information about which city is mentioned by which sources.

Overall, these sources have left me with 119 trade cities located in 10 European countries. The data set encompasses all 839 NUTS-3 regions in these countries.⁸ The Data Appendix offers a detailed

⁵ More information about the kind of information and the geographical and temporal scope of those maps is provided in the Data Appendix.

⁶ The cities of the Hanseatic League were assigned to three (later four) different regional subgroups, the “Thirds” (“Quarters”) according to their geographical location.

⁷ In Table B.1 I show that the results remain intact when I use alternative samples of trade cities, accounting for the uncertainty of the historical information on which the original coding is based.

⁸ I exclude the islands of Elba, Corsica and Sicily from the sample because they are not comparable with regions on the continent with respect to trade flows. Furthermore, I do not consider Spain or Denmark. The reason for this is that I am not able to find enough sources about trade routes and activities in these countries. Moreover, Spain was not a major player in medieval trade. It only become a leading trade nation in the early 16th century when over-seas trade began to rise. Finally, the institutional and political environment of Spain was considerably different from that in central Europe as parts of Spain were under Muslim rule until the later 15th century.

⁴ In Table B.2 in the Appendix, I account for the considerable differences in the size of NUTS-3 regions in various ways. First, I additionally include a country's average NUTS-3 region area as control variable. Second, I run regressions excluding Germany, Belgium and the Netherlands (the countries with much smaller NUTS-3 regions) and third, I run regressions inversely weighting the observations by their area (weighted least squares). And finally, I present estimates only considering regions that are comparatively homogeneous with respect to their size (i.e., have an area that is located within the 40th to 80th percentile of the region size distribution).

description of how the database of important late medieval trade cities is constructed.

Predominantly, I use three different variables as measures of late medieval trade and its impact on contemporary regional development. First, I use a dummy variable “Trade Center” that is equal to one if a region includes at least one medieval trade city. The lack of quantitative information and the limited availability of qualitative judgments led me to use a simple dummy variable coding important trade cities. Second, the theoretical reasoning implies that medieval trade affected city development through agglomeration. Because agglomeration is a process that takes place over centuries, cities that became an important trade center might therefore today be larger and richer on average because they were subject to agglomeration forces for a longer time. Thus, I create a variable that reports the number of centuries since a city in a region was recognized as an important trade city in my sources (see Data Appendix Table A.4).

Finally, the construction of a dummy variable also allows for the construction of a third variable “Distance to Trade Center” representing the distance (in degrees) between a region and the closest medieval trade city. This variable offers a useful direct test of my hypothesis that medieval trade contributed to the emergence of time persistent core-periphery patterns and therefore can act as an explanation for contemporary regional income differences.⁹

Table 1 provides a summary of the trade city data. For each country, the total number of NUTS-3 regions, the number of regions with trade cities, the share of trade center regions, the average distance of a region to the closest trade city and the average number of centuries since a region is coded as trade center is listed.

As reported in the table, the average distance to a medieval trade center is about 1.5° ($e^{0.432}$) which is approximately 170 km. Overall, around 14% of all regions are considered as containing medieval trade centers. On average, a trade city became an important trade center around 800 years ago (i.e. somewhere in the 14th century). Furthermore, Fig. 1 shows a map that depicts all included NUTS-3 regions and the regions with medieval trade centers (reddish colored).

Finally, with this data it is possible to take a look at the determinants of medieval trade. This is of interest because for empirical identification of the effect of medieval trade activities it is necessary to control for the determinants of medieval trade in general and especially those determinants that could also affect agglomeration. In Appendix D.1 I therefore run probit regressions where I make use of a city-level time-variant version of the medieval trade city variable (as it is used in Section 4.3) and the Bosker et al. (2013) data set to investigate the determinants of medieval trade.¹⁰ The results are reported in Table D.2. The results highlight the role of location

fundamentals (location on coast or river, elevation and agricultural productivity), of second nature geography (urban/ market potential, number of trade cities in the neighborhood of a city) and of archbishops and bishops as well as the existence of institutions of communal self-governance and the absence of a powerful territorial ruler. These insights serve as the basis for the further empirical analysis, in particular for the panel data analysis using the Bosker et al. (2013) data set later.

4. Empirical analysis

4.1. Medieval trade and contemporary regional development

4.1.1. Empirical specification

To test the main hypothesis that regions with cities involved in medieval trade exhibit higher levels of economic development today, I estimate the following equation using Ordinary Least Squares (OLS):

$$\ln(\text{GDP})_{cijk} = \alpha + \beta TC_{cijk} + \gamma'_1 X_{cijk} + \gamma'_2 X_{cij} + \theta_i + \lambda_j + \epsilon_{cijk} \quad (1)$$

where $\ln(\text{GDP})_{cijk}$ is the natural logarithm of GDP per capita in NUTS-3 region k NUTS-2 Region j in NUTS-1 region i of country c .

TC_{cijk} is a dummy variable “Trade Center” that is equal to one if a NUTS-3 region includes a medieval trade city and zero otherwise. Alternatively, TC_{cijk} represents a variable reporting the number of centuries since a trade city in a region became an important center of trade (“Centuries of Trade”). Finally, TC_{cijk} stands for the natural logarithm of the distance from a region's centroid to the closest trade city measured in degrees (“Distance to Trade Center”).¹¹ The latter variable allows to test whether medieval trade transmits its impact via agglomeration and concentration of economic activities. If this is true, a core-periphery pattern should emerge, with the medieval trade center regions as core and the other regions as periphery.

I expect β to be positively significant in the case of the “Trade Center” and the “Centuries of Trade” variable and negatively significant in the case of the “Distance to Trade Center” variable.

X_{cijk} and X_{cij} are vectors of NUTS-3 or NUTS-2 level covariates, controlling for the determinants of development and medieval trade, respectively. θ_i and λ_j are NUTS-1 and NUTS-2 region fixed effects. As NUTS-2 regions are perfect subsets of NUTS-1 regions I include either NUTS-1 or NUTS-2 region fixed effects. At last, ϵ_{cijk} is the error term capturing all unobserved factors. Eq. (1) is a straightforward way to establish a significant and direct link between late medieval trade activities and contemporary economic performance.

4.1.2. Controlling for determinants of medieval trade and regional development

To ensure that the significant positive relationship between medieval trade and contemporary economic development is not driven by omitted variables bias I control for relevant determinants of both medieval trade and economic development. The choice of the control variables is inspired by the insights about the determinants of medieval trade, as outlined in Section 3.2 and existing research on these topics. Since some of the control variables vary on NUTS-2 level I will regularly include NUTS-1 dummies instead of NUTS-2

⁹ Constructing the variable as distance to nearest trade route instead of distance to nearest trade city is not a viable alternative since on trade routes and smaller places along them no activity or economic transaction took place, they were simply transit routes leading from one commercial center to another. Furthermore, there seems to be a noticeable amount of uncertainty about the exact course of the trade routes. Hence, using distance to trade routes would possibly create another source of bias. In Table B.5, finally, I account for the fact that I do not include trade cities in some of the neighbors of the considered countries (e.g., I do not include Spain in the analysis, but France is included). For the border regions to these not included countries the distance variable might therefore be biased as the closest trade center is actually located in Spain and not in France (but I have not included cities in Spain). To overcome this issue, I delete the 50 regions that are located at a border to a country that is not included in the analysis and re-estimate the specifications in Table 2 Panel B.

¹⁰ To get a cross sectional data set which reflects the different centuries in which the cities became trade cities, I employed the following construction procedure to create the data set. Basically, I take the Bosker et al. (2013) data set and supplement it with time-variant versions of the three medieval trade variables. The final data set used for the regressions is then constructed as follows: First, for each city the century in which it became a trade city is recognized. Next, the values of the variables before this century are included in the data set to circumvent reverse causality issues. For non-trade cities the variable values in 1500 CE are considered.

¹¹ I use logarithmized values of the distance variable to get a regression coefficient that is an elasticity and therefore easy to interpret and comparable to the coefficient of the dummy variable. However, none of the results depend on this choice as the results would be virtually identical if I were instead to use the original distance values. Results not reported but available from the author upon request.

Table 1
The data on medieval trade centers.

Country	No. of regions	No. of trade centers	Share trade centers	Mean ln(Distance to Trade Center)	Mean centuries of trade
Austria	35	7	20	0.36	7.29
Belgium	44	3	6.8	0.41	7.67
Czech Republic	14	4	28.6	0.43	11.25
France	94	20	21.3	0.53	7.5
Germany	429	37	8.6	0.39	8.57
Hungary	20	2	10.0	0.69	6.5
Italy	90	25	27.8	0.41	7.46
Lithuania	7	2	28.6	0.56	6
Netherlands	40	7	17.5	0.29	6.29
Poland	66	12	18.18	0.55	6.58
Total	839	119	14.8	0.425	7.74

fixed effects. However, I will also run a specification with NUTS-2 dummies and exclude the variables defined at NUTS-2 level.¹²

At first, I add a set of basic geographic controls, including latitude, longitude and altitude of a NUTS-3 region. This set of variables should capture the general geographic pattern of development in central Europe. This means that economic development roughly increases from South to North (i.e., with increasing latitude) and decreases, in the sample, from West to East (i.e., with increasing longitude). Furthermore, it is widely acknowledged that regions with higher altitude are more difficult to reach—which seems especially relevant for trade—and have less favorable climates.

The second set of variables controls for the “Regional Environment and Location” of regions. It includes variables measuring the distance of a region to the closes physical geographic features that are important first nature determinants of agglomeration (coastlines and major rivers) and medieval trade (Börner and Severgnini, 2014; Bosker, 2015; Ellison and Glaeser, 1999 and Wolf, 2009).¹³ Additionally, the ln of the distance of each region to the contemporary border of a country is included.¹⁴ Furthermore, I include the average GDP per capita of regions within a 150-kilometer radius around the considered region (the GDP of the neighboring regions is inversely weighted by distance).¹⁵ Finally, this set of controls also incorporates four variables measuring the number of trade centers within four different distance bands (0–50 km, 50–100 km, 100–150 km and 150–250 km) around a city.¹⁶ These variables are intended to capture possible positive or negative spatial spillovers from nearby trade centers and the economic development of neighboring regions. Thus, they account for a region's second

nature geography and simultaneously allow me to directly control for spatial dependence.¹⁷

A third set of variables controls for relevant contemporary characteristics of the included regions. It comprises dummy variables for the regions that include a country's capital and a categorical variable identifying the degree to which a region may be considered a “mountain region” is also included. Furthermore, the set includes dummies for regions with coal or ore mines (or mining firms).¹⁸ Finally, it includes the ln of a region's area and the suitability of its soil for agriculture.

In consequence, the first two sets of controls account for important first (geographical circumstances) and second nature causes of agglomeration and medieval trade.

The next set of controls captures the historical characteristics of regions that could be relevant for both present day agglomeration and economic performance. Here I consider dummy variables indicating regions with a university founded before 1500 CE and regions that adopted printing technology before 1500 CE. As Cantoni and Yuchtman (2014), Dittmar (2011), and Rubin (2014) have demonstrated, both universities and printing technology are important factors in explaining the late medieval commercial revolution and city growth. What is more, I include a dummy variable reporting regions with a city acting as seat of a bishop or archbishop in 1500 CE. Furthermore, I also incorporate dummies for regions containing at least one imperial city, at least one city that was member of the Hanseatic League or a residence of a secular ruler.¹⁹ Finally, I also control for the possible long-lasting effect of a Roman Empire legacy and low transport costs for trade by including a dummy for cities located along an important imperial road and a variable reporting the distance of a city to the closest Roman road.

¹² In Appendix D.1 I present estimates of a baseline specification including only latitude, longitude, altitude and NUTS-1 or NUTS-2 dummies as controls. As Moran's I indicates the presence of spatial autocorrelation, there I also present standard errors estimated according to Conley's (1999) method. In general, those standard errors do not vary substantially from the heteroskedasticity robust version. In view of this, heteroskedasticity robust standard errors are reported in all remaining estimations.

¹³ In Table B.6 in the Appendix I additionally run regressions without coastal regions to ensure that e.g., the results are not driven by tourism.

¹⁴ These variables are also included to ensure that the distance to trade center variables does not actually capture the effect of possibly correlated distances to other important agglomeration factors, like e.g. rivers. I also run regressions including distance of a region's centroid to railroads and major roads. These are not shown because these variables are probably bad controls (Angrist and Pischke, 2009) but the results do not change when I include these additional variables. Results are available upon request.

¹⁵ I tested several other distance bands from 50 km to 2500 km. They are all highly correlated with each other and the results will not change substantially when these alternative measures are used.

¹⁶ As before I tried several additional distance bands (250–350 km and so on) and found that they are usually insignificant when added to the specification and have virtually no impact on the results).

¹⁷ Alternatively, I estimated regressions using a variable representing the sum of the inhabitants of all other cities apart from the considered one (again weighted by the inverse distance). This measure is analogously defined as the “urban potential” measure used to represent a city's second-nature geography (market potential, centrality in the urban network etc.) e.g., in Bosker et al. (2013). Using this variable leaves the coefficient of the trade variables virtually unchanged (results not shown but available from the author). In Tables B.3 and B.4 in the Appendix, I estimate the regressions including neighbor GDP per capita and urban potential measures (in the panel data case) using a spatial lag, a spatial error and a mixed spatial model. These are employed to show that spatial autocorrelation (whether explicitly modelled or in the error term) does not bias the estimates and also that simultaneity arising from the spatial lag variable does not lead to a distortion of the results.

¹⁸ This variable is also defined at NUTS-2 level and therefore only included in the specifications with NUTS-1 fixed effects.

¹⁹ The Hanseatic League was an association of German trading towns built for the purpose of, for instance, better ensuring the security of the traders. Later it became a political, economical and cultural alliance of the member cities that made decisions together about important economic and political matters on the so called “Hansetage”. Imperial cities were cities in the Holy Roman Empire that were almost independent political states and were politically subordinated directly to the Emperor and not the duke ruling the territory around them.

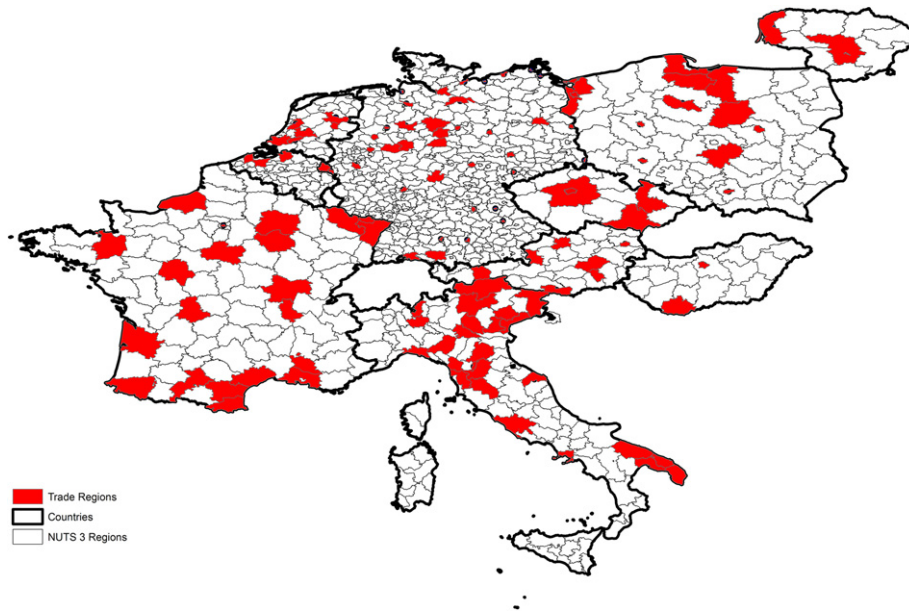


Fig. 1. NUTS-3 regions with medieval trade cities.

The fifth set controls for the most important covariates of economic growth and development. Here I use the share of people aged between 25 and 64 with tertiary education (on NUTS-2 level) as a measure for regional human capital. As a variable to measure the quality of regional economic and political institutions I use the quality of government index developed by the Quality of Government Institute at the University of Gothenburg, which provides a measure for regional institutional quality design similar to the World Governance Indicators (WGI) of the World Bank. To measure for regional inequality I construct the ratio of average workers compensation to GDP per capita. As measure of innovative activity within a region I use the number of patents registered by a region's firms at NUTS-2 level. Furthermore, I include a region's unemployment rate, $\ln$ of the average workers compensation, and the $\ln$ of the average fixed capital of a region's firm (all on NUTS-2 level).²⁰

The results of the regressions are shown in Table 2. First, I add the four sets of controls separately to the baseline specification and then I include all sets of controls jointly in one regression. I see that the coefficients of the trade center dummy and the distance variable remain significant in each of the specifications, although the sizes of the coefficients are reduced considerably when compared to the baseline estimates.

The coefficient is smallest (e.g., around 0.09 in the case of the trade center dummy) in the specification with all covariates added jointly to the baseline geographic controls and NUTS-1 dummies or, alternatively, when the variables defined on NUTS-2 levels are removed and instead NUTS-2 fixed effects are added. It suggests that medieval trade center regions today have a GDP per capita of some 10% higher than other regions. Based on the average regional GDP per capita in the sample this corresponds to a GDP per capita that is approximately 1900 Euros higher. Doing the same exercise for the

centuries since importance in trade variable, a region that is a trade center for the minimum number of centuries (i.e., since 1500 CE) has a GDP per capita that is around 7% higher ($[0.012 * 6] * 100$).

Overall, I see that the relationship between medieval trade and contemporary regional development is robust to the inclusion of a wide range of control variables and other important determinants of agglomeration and economic performance.

4.1.3. Accounting for unobserved heterogeneity

The biggest obstacle for the identification of the causal effect of medieval trade on contemporary economic development is unobserved heterogeneity. In particular, a third, unobserved fundamental factor of development could have influenced both the location of medieval trade centers as well as contemporary economic prosperity.

To further mitigate this possibility I will focus on a sub-sample of cities that were important urban centers around 1500 CE. The crucial idea here is that I will identify urban centers based on criteria like political importance or ecclesiastical importance. Hence, not all—but some—important urban centers will also be centers of trade. One can expect these cities to be a comparatively homogeneous group, in particular with respect to an unobserved historical factor of development responsible for their status as urban centers. If I found a significant positive effect of being a medieval trade center (i.e. not only being a political and/or ecclesiastical center but also a commercial center) this would provide evidence that the effect of medieval trade is not due to unobserved heterogeneity or captures a general developmental advantage of important urban centers.²¹

²⁰ When including these variables to the regression specification I lose several observations since they are not available or are not completely available for some countries. The unemployment rate for example is only available for 518 of the 839 regions. The inclusion of these controls might also cause a "bad controls problem". Thus, I only use them in one estimation and in all the other cases I present estimates without these control variables. Nevertheless, as most of them turned out to be significant it could be useful to explore how the coefficient of the medieval trade variables changes when considering them.

²¹ More ideally would be to have a valid instrumental variable for medieval trade activities. However, due to wide spatial coverage of the sample, the fact that medieval trade developed over several centuries with shifting gravitational centers to find such an instrument is not an easy task. However, in an early version of the paper (Wahl, 2013) I present 2SLS estimations using mountain regions and historical bishop seats as instruments. There, I also present results using Lewbel's (2012) approach with artificial instruments. In the Appendix E to this paper I include a revised version of the 2SLS estimations using historical bishops seats as instruments, although one can argue about whether the exclusion restriction holds in this case. The results indicate that the OLS estimates are actually downward biased and show an even larger positive effect of medieval trade. Appendix D.5 also provides a discussion of certain other possibilities to cope with the endogeneity issue and their problems in the context of the paper.

Table 2
Medieval trade and contemporary regional economic development.

Dep. var.	ln(GDP per capita)					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: trade center dummy</i>						
Trade center dummy	0.222*** (0.024)	0.167*** (0.021)	0.178*** (0.029)	0.130*** (0.021)	0.0938*** (0.019)	0.105*** (0.025)
Adj. R ²	0.787	0.843	0.79	0.878	0.904	0.853
<i>Panel B: distance to nearest trade center</i>						
ln(Distance to Trade Center)	−0.243*** (0.038)	−0.125*** (0.32)	−0.128*** (0.04)	−0.138*** (0.0317)	−0.106*** (0.029)	−0.107*** (0.041)
Adj. R ²	0.777	0.834	0.782	0.872	0.902	0.852
<i>Panel C: centuries of trade</i>						
Centuries of trade	0.0285*** (0.003)	0.0212*** (0.003)	0.0235*** (0.004)	0.0169*** (0.003)	0.012*** (0.002)	0.0131*** (0.003)
NUTS-1 dummies	Yes	Yes	Yes	Yes	Yes	No
NUTS-2 dummies	No	No	No	No	No	Yes
Basic geographic controls	Yes	Yes	Yes	Yes	Yes	Yes
Regional environment and location	Yes	No	No	No	Yes	Yes
Region characteristics	No	Yes	No	No	Yes	Yes
Historical region characteristics	No	No	Yes	No	Yes	Yes
Growth covariates	No	No	No	Yes	Yes	No
Obs.	839	839	839	518	518	839
Adj. R ²	0.789	0.843	0.791	0.878	0.904	0.853

Notes. Robust standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1 %, **5 % and *10 % level. The unit of observation is a NUTS-3 region. The basic geographic controls include a region's latitude, longitude and altitude. The "Regional Environment and Location" controls include the ln distances of a region's centroid to the nearest border, coast point or river as well as the distance weighted sum of the GDP per capita of all NUTS-3 region in a radius around 150 km around the considered NUTS-3 region and four variables indicating the number of medieval trade regions 0–50, 50–100, 100–150 and 150–250 km away. Region characteristic controls include dummies for regions including a country's capital, are classified as mountain regions and with ore or coal mines (this latter variable is not included in column (6) as it would be collinear with the NUTS-2 fixed effects). Furthermore, it encompasses the ln of a regions area and the ln of a regions agricultural suitability. The historical region characteristics consist of a dummy variables indicating regions with a city serving as seat of a bishop in 1500 CE, with a university founded before 1500 CE, that adopted printing technology before 1500 CE, contain cities that were members of the Hanseatic League, with former imperial cities, that were residence cities of a secular ruler or were located on an Imperial road. Moreover it includes the ln of the distance of a regions centroid to the closest Roman road. The growth covariates encompass a region's unemployment rate, number of registered patents, average firm ln fixed capital stock, average worker compensation. Furthermore, it includes the share of people aged between 25–64 with tertiary education on NUTS-2 level, the quality of government index on NUTS-1/ NUTS-2 level and the ratio of an average workers compensation to a region's GDP per capita as inequality measure. Each regression includes a constant not reported.

To build the sample of historical urban centers I first construct an urban centrality index in the spirit of [Escher and Hirschmann \(2005\)](#). I create the index by simply summing up seven proxy variables for the strategic, political or administrative and ecclesiastical importance of a region in the later medieval period (i.e., if a city is of strategic importance). The selection is based on insights from existing studies on the determinants of city development and the determinants of urban centrality (e.g., [Bosker et al., 2013](#); [Bosker, 2015](#); [Cantoni and Yuchtman, 2014](#); [Dittmar, 2011](#); [Escher and Hirschmann, 2005](#) and [Roos, 2005](#)). I do not include proxies for the economic or commercial importance of the city as cities that were important ecclesiastical and or political centers but not commercial centers constitute my control group. Those seven variables are a dummy variable reporting the presence of a bishop or archbishop in 1500 CE as proxy for ecclesiastical importance; dummy variables reporting imperial and residence cities around 1500 CE as proxies for political and strategic importance; and the ln of a region's centroid to the closest major river or Roman road and a dummy equal to one in regions with access to the sea representing a city's geographic fundamentals, strategic importance and Roman heritage.²² The resulting index is significantly positively correlated with contemporary population density and relative GDP density and GDP per capita.²³ This once again documents the striking persistence of urban centrality

over centuries that contrasts the "Reversal of Fortune" experience of the colonized countries as analyzed by [Acemoglu et al. \(2002\)](#). If I still find a significant and persistent effect of medieval trade for this group of cities it can probably be attributed to medieval trade and not to other causes of persistence of urban centrality.

In the next step, I re-estimate the specifications in [Table 2](#) column (6) for each of the three medieval trade measures and three different sub-samples. The results are reported in [Table 3](#). In columns (1)–(3) of [Table 3](#) I report the results when using the sub-sample of regions with an above average index of urban centrality. In columns (4)–(6) I only consider regions in the 4th quantile of the urban centrality index and in columns (7)–(9) finally I run the regressions on the regions in the 90th percentile of the urban centrality index, i.e. the 83 regions with the highest urban centrality scores. In the latter case, it is not possible to include all the control variables used in [Table 2](#) column (6) due to the substantially lower number of observations. To safe degrees of freedom I therefore include NUTS-1 fixed effects instead of NUTS-2 fixed effects and only three additional control variables (district-free city and capital dummies as well as the agricultural suitability measure).²⁴

Regardless of which sub-sample is used, the three medieval trade measures remain significant. Furthermore, they actually show larger coefficient values implying that if unobserved heterogeneity biased the estimated effect of medieval trade it downward biased it. Nevertheless, all the coefficients remain within the range of [Table 2](#)

²² The distance variables are rescaled so that they are positively correlated with urban centrality and take values between 0 and 1. Hence, the index can have a maximum value of seven and a minimum value of zero. However, the actual mean of the index is 1.066, with a minimum of 0.20, a maximum of 4.438 and a standard deviation of 0.702.

²³ The correlations with relative GDP and population density are 0.17 and 0.18, respectively. Additionally, the correlation with regional GDP per capita is around 0.21

²⁴ These additional controls are selected based on the following procedure: First, I include each set of control variables separately to the NUTS-1 dummies. Next, I include all the variables that were significant in the first step jointly. Finally, I retain only the variables that remain significant in step two, which are those three variables.

Table 3

Medieval trade and contemporary economic development—historical urban centers as control group.

Dep. var.	ln(GDP per capita)								
	Urban centrality index $\bar{z}$ Average			Urban centrality index 4th quantile			Urban centrality index 90th percentile		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Trade center	0.118*** (0.04)			0.174*** (0.038)			0.181*** (0.063)		
ln(Distance to Trade Center)		−0.161** (0.072)			−0.244*** (0.076)			−0.197** (0.097)	
Centuries of trade			0.0147*** (0.005)			0.0202*** (0.005)			0.0216*** (0.006)
NUTS-1 dummies	No	No	No	No	No	No	Yes	Yes	Yes
NUTS-2 dummies	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
Basic geographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
City environment and location	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
Region characteristics	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
Historical region characteristics	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
Growth covariates	No	No	No	No	No	No	No	No	No
Robust controls	No	No	No	No	No	No	Yes	Yes	Yes
Obs.	338	338	338	210	210	210	84	84	84
Adj. R^2	0.855	0.854	0.856	0.85	0.845	0.849	0.848	0.833	0.859

Notes. Robust standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1 %, **5 % and *10 % level. The unit of observation is a NUTS-3 region. The basic geographic controls include a region's latitude, longitude and altitude. The "Regional Environment and Location" controls include the ln distances of a region's centroid to the nearest border, coast point or river as well as the distance weighted sum of the GDP per capita of all NUTS-3 region in a radius around 150 km around the considered NUTS-3 region and four variables indicating the number of medieval trade regions 0–50, 50–100, 100–150 and 150–250 km away. Region characteristic controls include dummies for regions including a country's capital and are classified as mountain regions. Furthermore, it encompasses the ln of a regions area and the ln of a regions agricultural suitability. The historical region characteristics consist of a dummy variables indicating regions with a city serving as seat of a bishop in 1500 CE, with a university founded before 1500 CE, that adopted printing technology before 1500 CE, contain cities that were members of the Hanseatic League, with former imperial cities, that were residence cities of a secular ruler or were located on an Imperial road. Moreover it includes the ln of the distance of a regions centroid to the closest Roman road. The set of robust controls is selected according to the procedure described in the text. It comprises of the capital dummy, the ln of a region's area and the number of trade cities 50–100 km around a city. Each regression includes a constant not reported.

and especially the coefficient of the centuries of trade variable turns out to be stable across different samples and specifications.

All in all, these results provide suggestive evidence that even within a relatively small group of historical urban centers, those urban centers that were also commercial centers had an additional and persistent development advantage. It is probable that this advantage is not caused by one or more unobserved historical factors making some places "big and rich" and others "small and poor" but actually by medieval trade activities.

4.2. Medieval trade and contemporary agglomeration and industry concentration patterns

The next step is to establish the relationship between medieval trade and contemporary economic agglomeration.

I will achieve this by conducting regressions relating the medieval trade variables to measures of present-day agglomeration and industry concentration. Namely, I will estimate the following cross-sectional equation using OLS:

$$AGG_{cijk} = \alpha + \beta TRADE_{cijk} + \gamma'_1 X_{cijk} + \gamma'_2 X_{cij} + \theta_i + \lambda_j + \epsilon_{cijk} \quad (2)$$

where AGG_{cijk} represents the agglomeration measures, i.e. either the ln of the relative GDP density of a region or the ln of its population density. $TRADE_{cijk}$ represents one of the three medieval trade measures and the remaining terms of the equation are identically defined to Eqs. (1) and (2).²⁵ The results are presented in Table 4.

I see that all three medieval trade measures are highly significant and show the right signs in both the case of population density and for relative GDP density. The reported coefficients are large implying economic relevance of the estimated statistically significant effects. The results indicate that, as proposed by my theoretical hypotheses,

medieval trade is positively associated with contemporary agglomeration and industry concentration measures.

In Table 5 I additionally see that the significant impact of medieval trade on agglomeration exists even when I consider only the subsample of urban centers in 1500 CE, i.e., only cities in the 90th percentile of the Urban Centrality Index. Hence, even in the group of important urban centers, those urban centers that additionally were prominently involved in medieval trade show a higher degree of agglomeration today. This makes it more unlikely that the results are driven by unobserved heterogeneity and primarily reflect unobserved differences between large cities and the countryside. In fact, I found even larger effects indicating that whatever speculative unobserved heterogeneity exists in the overall sample it downward rather than upward biases the estimated effect of medieval trade.

In Appendix D.3 I additionally conduct a mediation analysis using night light intensity as proxy for regional GDP per capita and population density as variable mediating the relationship between medieval trade and present days economic development. Because night light intensity is not mechanically correlated with the agglomeration measures—as GDP per capita is—the use of this variable allows to directly relate the agglomeration measures to a proxy for economic development. Moreover, this provides a robustness check for whether the results are sensitive to a different measure of economic development. The results of the mediation analysis fully support my theoretical reasoning and show that the effect of medieval trade on economic development is significantly mediated by its influence on agglomeration patterns.

4.3. Medieval trade and city development—panel data evidence

Until now I found evidence that medieval trade activities significantly affect today's regional GDP per capita via their impact on agglomeration patterns. However, the observed relationship could be incidental. To show a persistent effect of medieval trade explicitly leading to a path-dependent regional development process I need to document that the positive relationship exists over a longer

²⁵ As there is a mechanical correlation between a region's area and both its population and relative GDP density I do not control for a region's area in these regressions.

Table 4
Medieval trade and contemporary agglomeration patterns.

Dep. var.	ln(Population Density)			ln(Relative GDP Density)		
	(1)	(2)	(3)	(4)	(5)	(6)
Trade center	0.514*** (0.124)			0.924*** (0.278)		
ln(Distance to Trade Center)		−0.669*** (0.164)			−1.261*** (0.367)	
Centuries of trade			0.0666*** (0.015)			0.118*** (0.033)
NUTS-2 dummies	Yes	Yes	Yes	Yes	Yes	Yes
Basic geographic controls	Yes	Yes	Yes	Yes	Yes	Yes
City environment and location	Yes	Yes	Yes	Yes	Yes	Yes
Region characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Historical region characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	839	839	839	839	839	839
R ²	0.596	0.593	0.598	0.431	0.429	0.432

Notes. Robust standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1 %, **5 % and *10 % level. The unit of observation is a NUTS-3 region. The “Regional Environment and Location” controls include the ln distances of a region's centroid to the nearest border, coast point or river as well as the distance weighted sum of the GDP per capita of all NUTS-3 region in a radius around 150 km around the considered NUTS-3 region and four variables indicating the number of medieval trade regions 0–50, 50–100, 100–150 and 150–250 km away. Region characteristic controls include a dummies for regions including a country's capital and are classified as mountain regions. Furthermore, it encompasses the ln of a region's agricultural suitability. The historical region characteristics consist of a dummy variables indicating regions with a city serving as seat of a bishop in 1500 CE, with a university founded before 1500 CE, that adopted printing technology before 1500 CE, contain cities that were members of the Hanseatic League, with former imperial cities, that were residence cities of a secular ruler or were located on an Imperial road. Moreover it includes the ln of the distance of a regions centroid to the closest Roman road. Each regression includes a constant not reported.

Table 5
Medieval trade and contemporary agglomeration patterns—subsample of historical urban centers.

Dep. var.	ln(Population Density)			ln(Relative GDP Density)		
	(1)	(2)	(3)	(4)	(5)	(6)
Trade center	0.632*** (0.172)			0.813*** (0.184)		
ln(Distance to Trade Center)		−1.035*** (0.278)			−1.232*** (0.304)	
Centuries of trade			0.0727*** (0.018)			0.0943*** (0.018)
NUTS-1 dummies	Yes	Yes	Yes	Yes	Yes	Yes
Robust controls	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	83	83	83	83	83	83
R ²	0.855	0.866	0.867	0.954	0.955	0.86

Notes. Robust standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1 %, **5 % and *10 % level. The unit of observation is a NUTS-3 region. The set of robust controls is selected according to the procedure described in the text. It comprises of the capital dummy, the ln of a region's area and the number of trade cities 50–100 km around a city. Each regression includes a constant not reported.

time period. Thus, I have to show the positive relationship between medieval trade and economic development in a panel data set. This also enables me to estimate the effect of trade more precisely since I can allow for the fact that some cities became important trade centers earlier or later in a more direct way than with the centuries of trade variable. Furthermore, it allows me to show that the relationship actually existed in the medieval period itself.

Hence, I run regressions using the Bosker et al. (2013) city level panel data set supplemented by time-varying versions of the three measures of medieval trade. The Bosker et al. (2013) data set spans the period from 800–1800 CE and provides data for every one-hundred years (i.e., 800, 900, 1000, 1100 etc.) and European cities for which Bairoch et al. (1988) provide city figures. I include every city in the Bosker et al. (2013) data set that is located in one of the ten countries I considered in the previous analysis. This leaves me with a sample of 362 cities of which 91 are coded as trade cities.²⁶ This leads to 3982 city-century pairs and—as city population data is

not available for each period in many cities—an unbalanced panel of 1533 observations will be used for the subsequent empirical analysis.

To uncover the long-run relationship between medieval trade and city development in a first step I estimate the following equation:

$$\ln(\text{POP})_{cit} = \alpha + \rho \text{TRADE}_{cit} + \gamma_1' \mathbf{X}_{cit} + \gamma_2' \mathbf{X}_{ci} + \theta_c + \lambda_t + \epsilon_{cijk} \quad (3)$$

where $\ln(\text{POP})_{cit}$ is the natural logarithm of a population of city i in country c in year t (with t being 800, 900, 1000...). TRADE_{cit} represents time variant versions of the three measures of medieval trade. $\mathbf{X}_{cit}$ is a set of time-varying control variables. Among them are dummy variables indicating whether a city was plundered in the previous century, had a university, held the status of a capital of a territorial state, was located in or surrounded by a state with a large territory (i.e., no city state), was located in a territory ruled by a non-absolutist monarch, was residence of a bishop or archbishop and a variable representing the presence of a local participative government (dummy variable equal to one if a city had a city council and zero if not). Furthermore, a variable of a city's urban potential according to the definition of DeVries (1984) is used. This variable represents the distance weighted sum of the size of all Christian cities around a particular city additionally taking into account the

²⁶ The Data Appendix reports every city coded as trade city and also reports the date since it is considered as important trade center. Table A.2 in the Data Appendix provides a descriptive overview of the variables employed in these panel data estimations.

Table 6
Medieval trade and city development—panel data estimations.

Dep. var.	ln(Population)					
	(1)	(2)	(3)	(4)	(5)	(6)
Method	RE			FE		
Period	800–1800	800–1500	1600–1800	800–1800	1st lag	2nd lag
<i>Panel A</i>						
Trade city	0.331*** (0.056)	0.269*** (0.065)	0.383*** (0.075)	0.230** (0.094)	0.261*** (0.08)	0.199*** (0.076)
R ²	0.453	0.426	0.523	0.368	0.369	0.358
<i>Panel B</i>						
ln(Distance to Trade City)	−0.0718*** (0.012)	−0.0553*** (0.014)	−0.0867*** (0.018)	−0.0477** (0.02)	−0.0524*** (0.017)	−0.0329** (0.016)
R ²	0.45	0.424	0.518	0.368	0.367	0.355
<i>Panel C</i>						
Centuries of trade	0.0581*** (0.011)	0.0734*** (0.017)	0.0486*** (0.01)	0.0514** (0.021)	0.0518** (0.022)	0.0458** (0.022)
Country dummies	Yes	Yes	Yes	No	No	No
Century dummies	Yes	Yes	Yes	Yes	Yes	Yes
City dummies	No	No	No	Yes	Yes	Yes
Geography controls	Yes	Yes	Yes	No	No	No
City characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	1533	709	824	1533	1512	1488
R ²	0.457	0.432	0.522	0.372	0.366	0.356

Notes. Standard errors clustered on city level in parentheses. Coefficient is statistically different from zero at the ***1 %, **5 % and *10 % level. The unit of observation is a city. The geographic controls include a city's latitude, longitude and altitude, the standard deviation of elevation in the region 10 km around the city, as well as dummy variables indicating location at a navigable river, sea, Roman road or hub of a Roman road. Additionally, those controls include a variable indicating whether a city belongs to one of four different ecozones (see Bosker et al., 2013 for more details). City characteristics incorporate dummy variables indicating whether a city was plundered in the previous century, harbors a university, had the status of a capital, was located in a state with a large territory and was located in a territory ruled by a non-absolutist monarch. Furthermore, they include a city's urban potential according to the definition of DeVries (1984) and three variables capturing the number of other trade cities 0–50 km, 50–100 km and 100–150 km around a city. In the case of random effects estimation (column (1)–(3)) the overall R² and in the case of the fixed effects estimation the within R² is reported. Each regression includes a constant not reported.

accessibility of a city by sea or navigable river. The variable accounts for the urban environment in which a city is embedded. All these variables are taken from the Bosker et al. (2013) city-level data set. Three supplementary variables are included capturing the number of trade cities 0–50 km, 50–100 km and 100–150 km around a city. Both the urban potential measure and the number of neighboring trade cities capture the second-nature geography of a city and spatial spillovers arising from nearby trade activities. $\mathbf{X}_{ci}$ is a vector of time-invariant control variables primarily encompassing geographic and biogeographic features of a city and the surrounding region (i.e., first-nature geography). Those variables include a city's latitude, longitude and altitude, the standard deviation of elevation in the region 10 km around the city, as well as dummy variables indicating location at a navigable river, sea, Roman road or hub of a Roman road. Additionally, a variable indicating whether a city belongs to one of four different ecozones is included (see Bosker et al., 2013 for more details). Those controls are chosen because they are correlated with both city development and medieval trade (Bosker et al., 2013; Cantoni and Yuchtman, 2014; Börner and Severgnini, 2014 among others). θ_c are country fixed effects and λ_t are century fixed effects absorbing temporal shocks which affected all countries (such as the Black Death). I also estimate a variant of Eq. (3) replacing the country fixed effects by city fixed effects (δ_i) to account for all time-invariant unobserved heterogeneity. All control variables and the city population figures originate from the Bosker et al. (2013) data set.

I estimate Eq. (3) for the whole sample period from 800–1800, for the medieval (800–1500) and the early modern period (1600–1800) separately. This tests whether the effect of medieval trade differs remarkably between different historical eras and whether there is an effect of trade on city development during the medieval era. If the latter is not the case, I cannot in fact attribute the effect of the medieval trade variables to medieval trade activities but to something else.

Furthermore, I estimate Eq. (3) for the whole observation period but using first and second order lags of the trade variables to account for reverse causality problems.

The results of these regressions are shown in Table 6.

All three trade measures are statistically and economically significant in all estimations. The coefficient in column (4), i.e. the FE regression using contemporary values of the trade city dummy implies that throughout the complete sample period, trade cities on average had a population about 26 % larger than non-trade cities. This effect is larger than in the cross-sectional estimations conducted in the previous sections. However, the obtained coefficients are more or less identical regardless of whether random or fixed effects are used for the estimation or which observation period is considered. The coefficient of the first order lags is larger than that of the contemporary values. However, the coefficient of the second order lags is lower. Since there are 200 years between the values of the medieval trade variable and the city population this is nevertheless a strong effect. These conclusions also hold true for the other two trade variables although the negative effect of distance to a trade city is lower than in the cross-sectional analysis.

The results are in line with a persistent effect of medieval trade on city development and they additionally show that trade did influence city development in the medieval period.

To further illustrate the persistent temporal effect of medieval trade on city development, I interact the trade city dummy with the century dummies and re-estimate the FE specification in column (4) of Table 6 including the interaction terms. The results of this are shown in Appendix D.4, Table D.5. The coefficients of the interaction terms imply a meaningful temporal evolution pattern of the coefficient of the trade center dummy confirming the hypotheses of a persistent effect of medieval trade, an effect already present during the medieval period and an insignificant effect before.

5. Conclusion

This paper argues that medieval trade led to agglomeration and a concentration of economic activities within the region it took place. It further postulates that the observed spatial distribution of population and economic activity across Europe today is still shaped by the self-reinforcing and long-lasting agglomeration processes that have their origins in medieval trade activities.

Empirical tests of these hypotheses confirmed, as expected, that there is a statistically and economically significant positive relationship between medieval trade activities and contemporary regional economic development. The analysis further provides evidence that this relationship is indeed caused by the influence that medieval trade exerted on the emerging patterns of agglomeration and spatial concentration of industrial activities throughout European regions. Based on the result of this paper, I found evidence for a causal chain running from medieval trade activities through medieval city growth to contemporary industry concentration and regional economic development. Medieval trade can therefore be considered as an important determinant of modern economic development and long-run city development. Further quantitative analyses of medieval trade activities based on detailed historical data will clearly improve the understanding of the sources of long-lasting economic and social prosperity—a subject that is of interest to researchers in a number of academic fields.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.regsciurbeco.2016.06.011>.

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